

YOLO Model Analysis Report

Model:	yolov8m_finetuned_1
Model Size:	49.6 MB
Parameters:	25.9 M
FLOPs (640x640):	79.09 GFLOPs
Run Name:	yolov8m_finetuned_1_testing_20251127_124030
W&B Run Name:	yolov8m_finetuned_1_bdd100k_yolo_limited_test_20251127_124030
Timestamp:	2025-11-27 12:52:01
Dataset:	bdd100k_yolo_limited - test split
Images Processed:	20000
Total Execution Time	118.54s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	2.37 ms per image
FPS (Frames Per Second)	421.62

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.6972
Recall	0.7814
F1-Score	0.7369
mAP@0.5	0.5683

mAP@0.5:0.95	0.3236
True Positives	256382
False Positives	111345
False Negatives	71716

Confusion Matrix

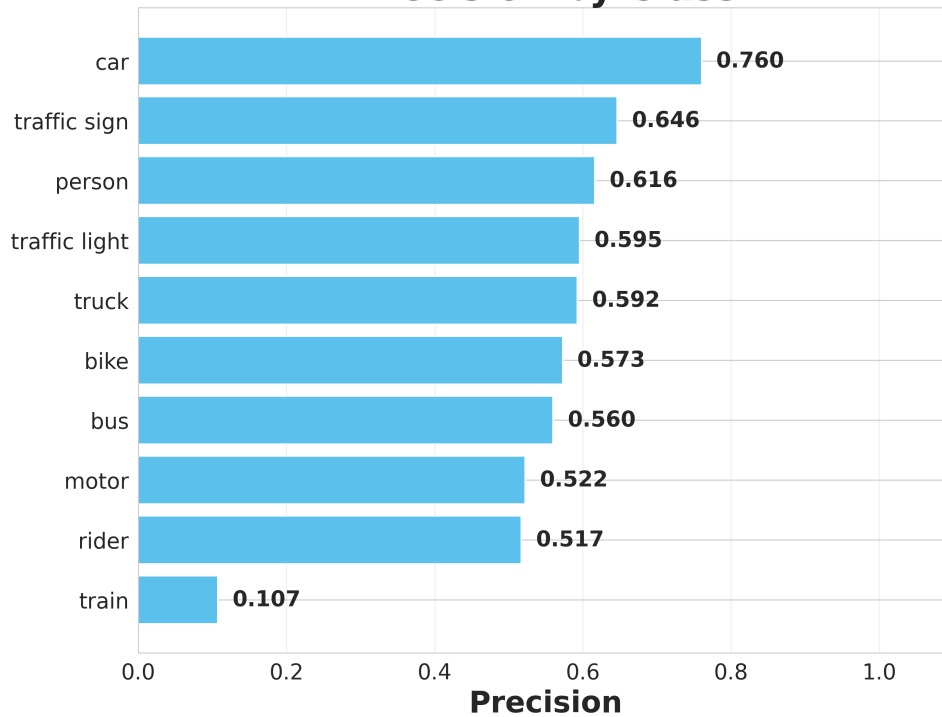
Confusion Matrix (yolov8m_finetuned_1 validation)

True Class	person	rider	car	truck	bus	train	motor	bike	traffic light	traffic sign
	15187	156	104	2			7	24	6	12
	215	669	9				7	8		
	114	1315	5933	1373	321		38	10	29	91
	2		993	5157	386	1			1	30
			286	427	1801	7				3
						3				
	11	4	38				439	34		1
	30	6	4			1	61	1144		
	9	1	23	2	2	3	1	2	31460	394
	29	1	89	14	5		2	1	281	44589
Predicted Class										

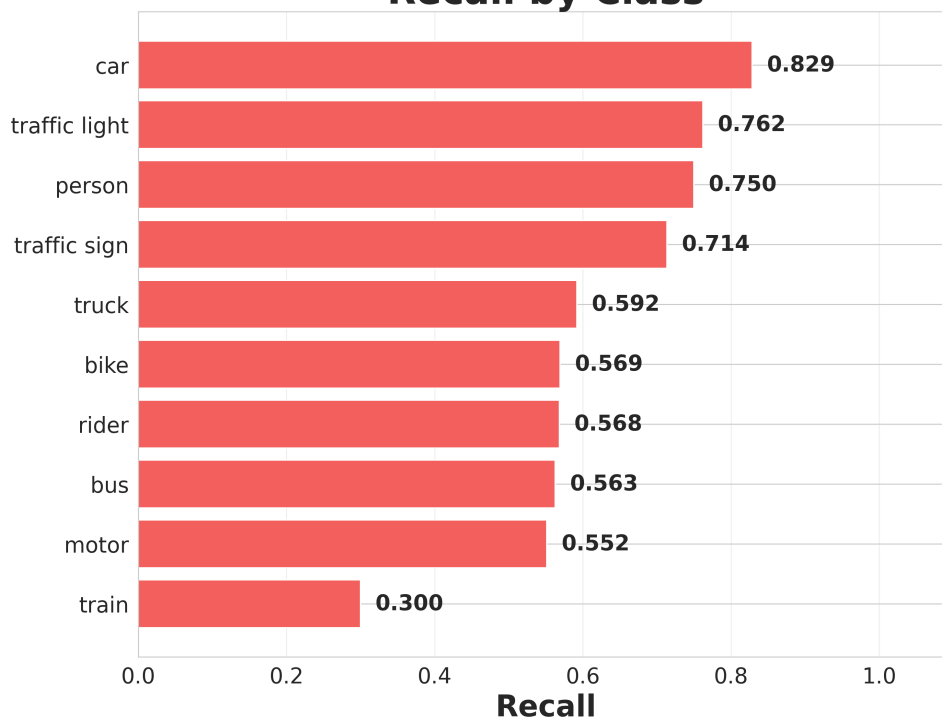
Correct Predictions (Diagonal Sum): 256382
Total Matched Predictions: 433719

Performance Visualizations

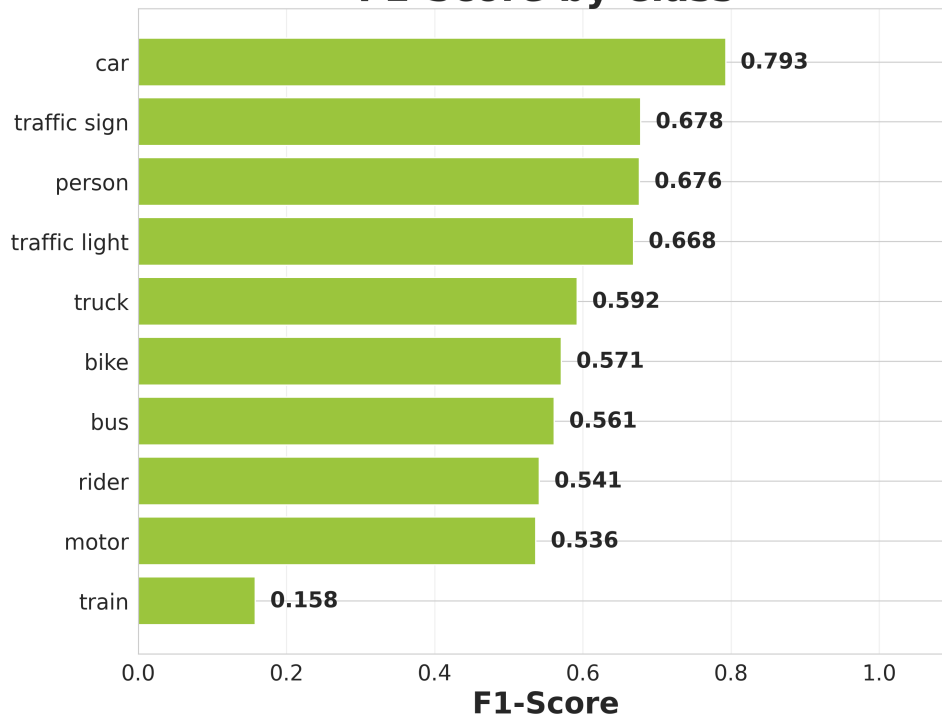
Precision by Class



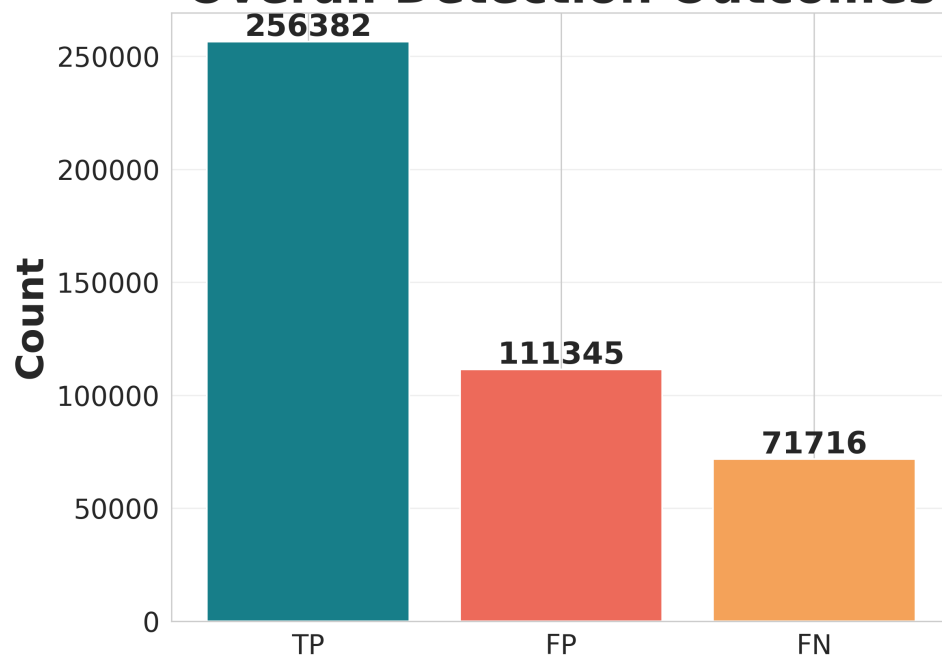
Recall by Class

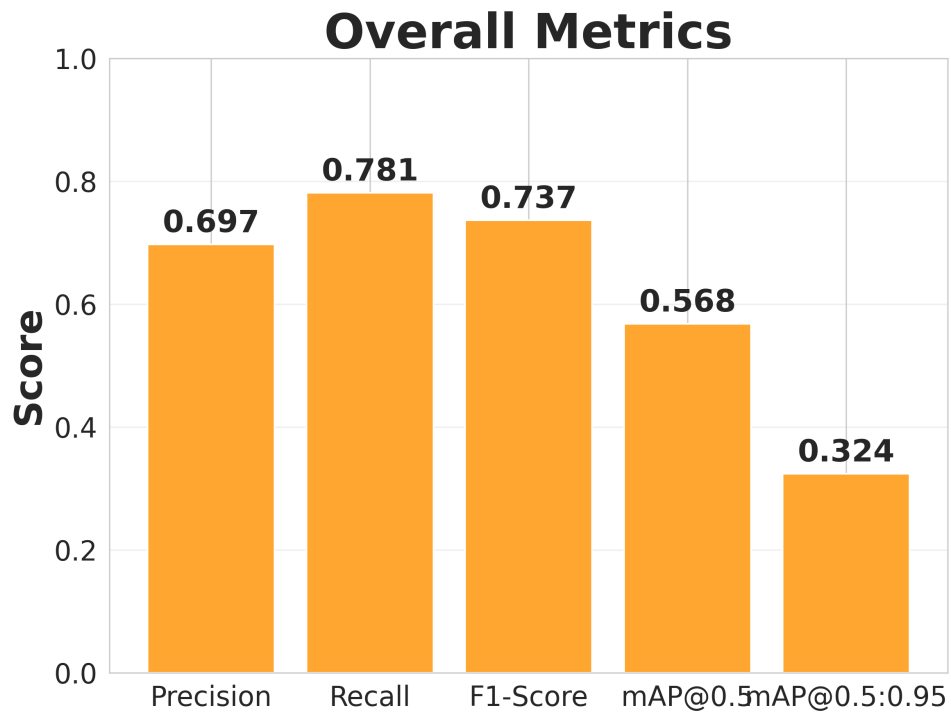
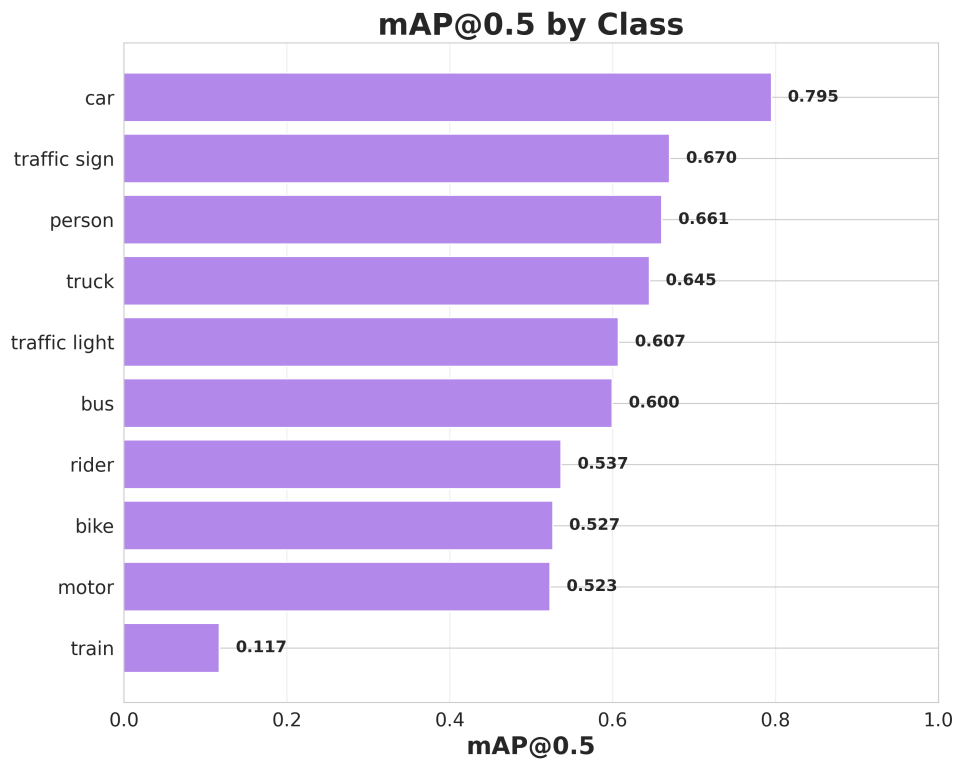


F1-Score by Class



Overall Detection Outcomes





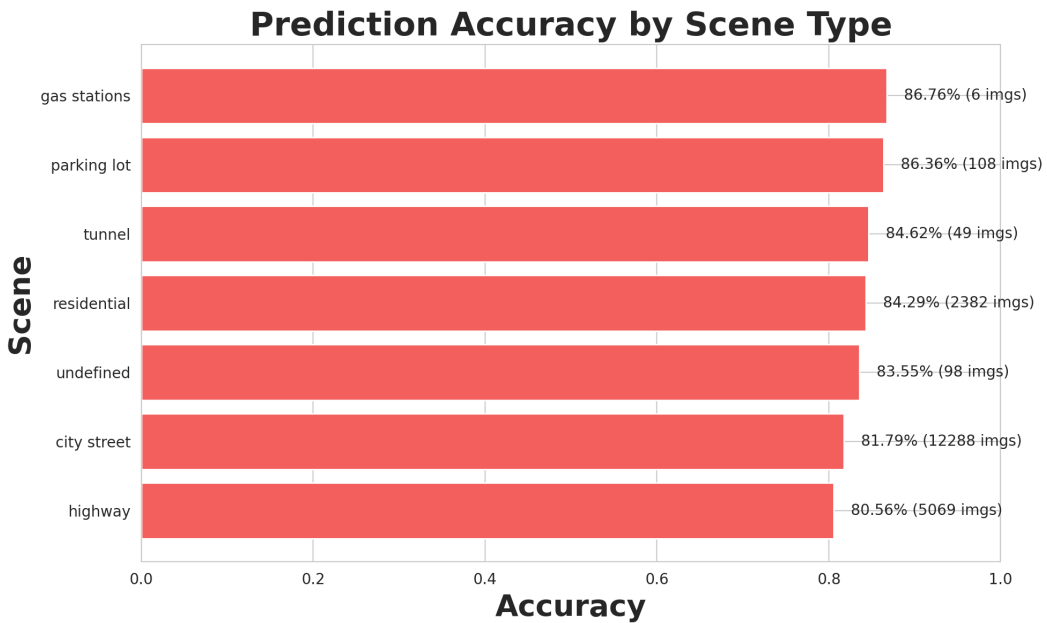
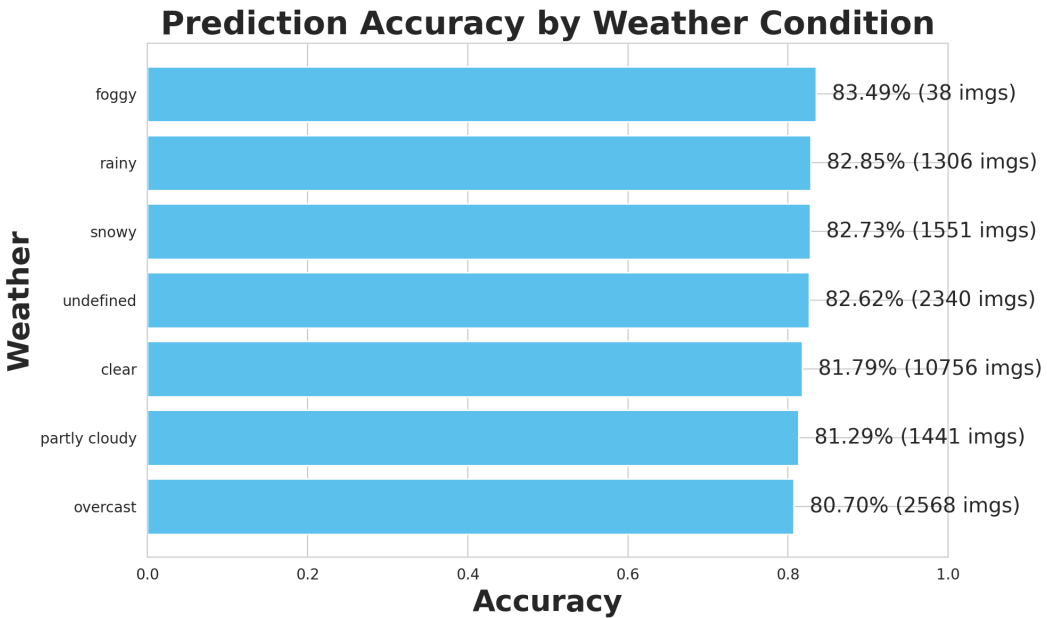
Per-Class Performance

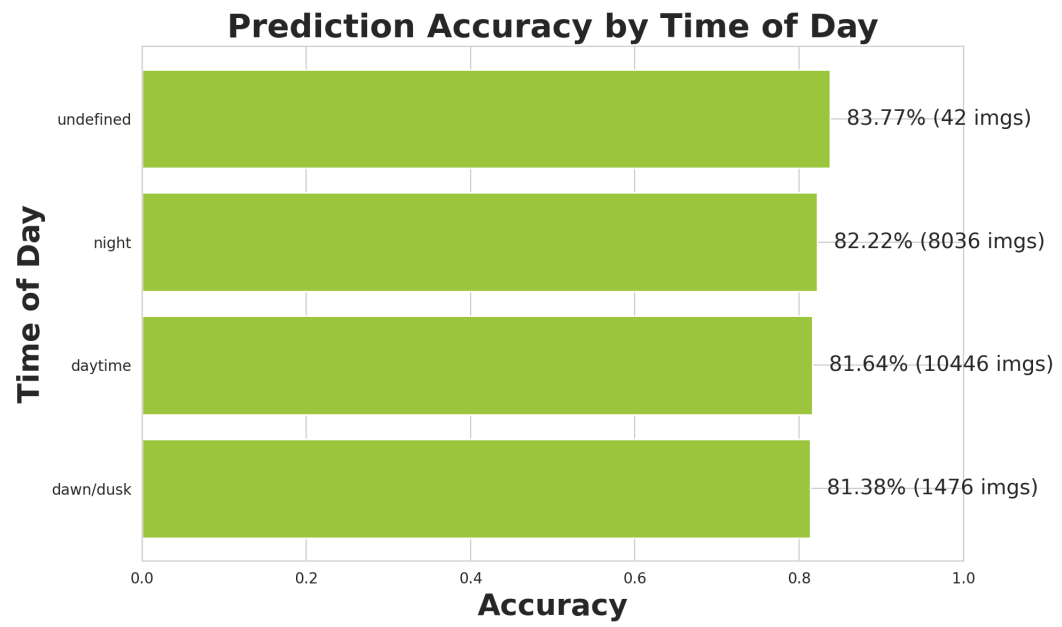
Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
person	15187	9463	5071	0.6161	0.7497	0.6764	0.6605
rider	669	625	508	0.5170	0.5684	0.5415	0.5368
car	155933	49216	32234	0.7601	0.8287	0.7929	0.7951
truck	5157	3547	3553	0.5925	0.5921	0.5923	0.6452
bus	1801	1416	1399	0.5598	0.5628	0.5613	0.5998
train	3	25	7	0.1071	0.3000	0.1579	0.1174
motor	439	402	357	0.5220	0.5515	0.5363	0.5235
bike	1144	854	865	0.5726	0.5694	0.5710	0.5267
traffic light	31460	21380	9822	0.5954	0.7621	0.6685	0.6072
traffic sign	44589	24417	17900	0.6462	0.7135	0.6782	0.6702

Comprehensive Failure Analysis

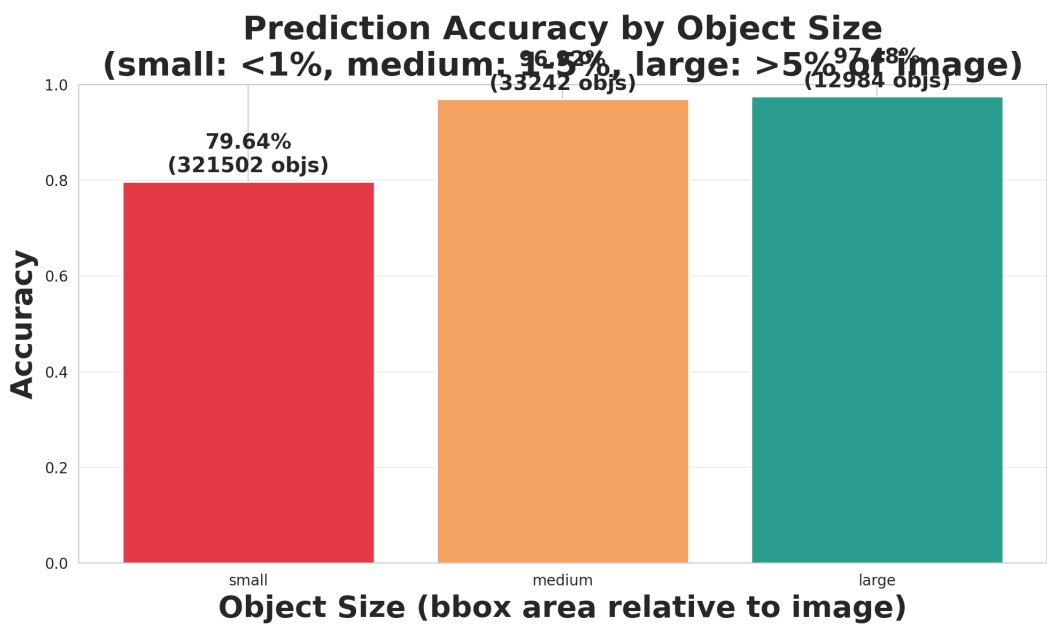
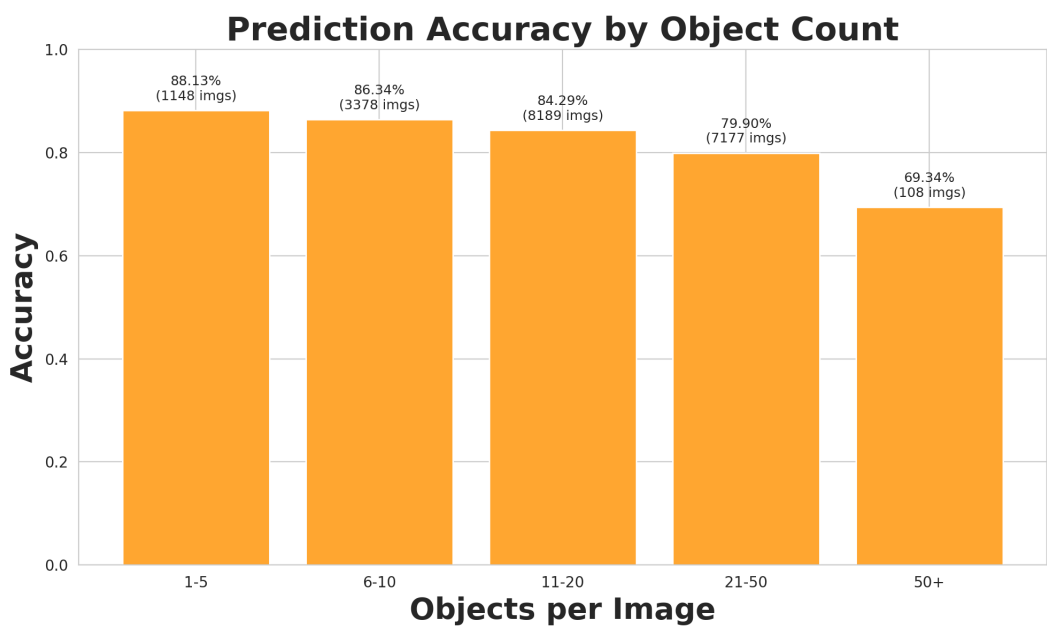
Metric	Value
Total Images Analyzed	20000
Total Expected Objects	367728
Total Matched Objects	300926
Overall Accuracy	81.83%

Accuracy by Environmental Attributes

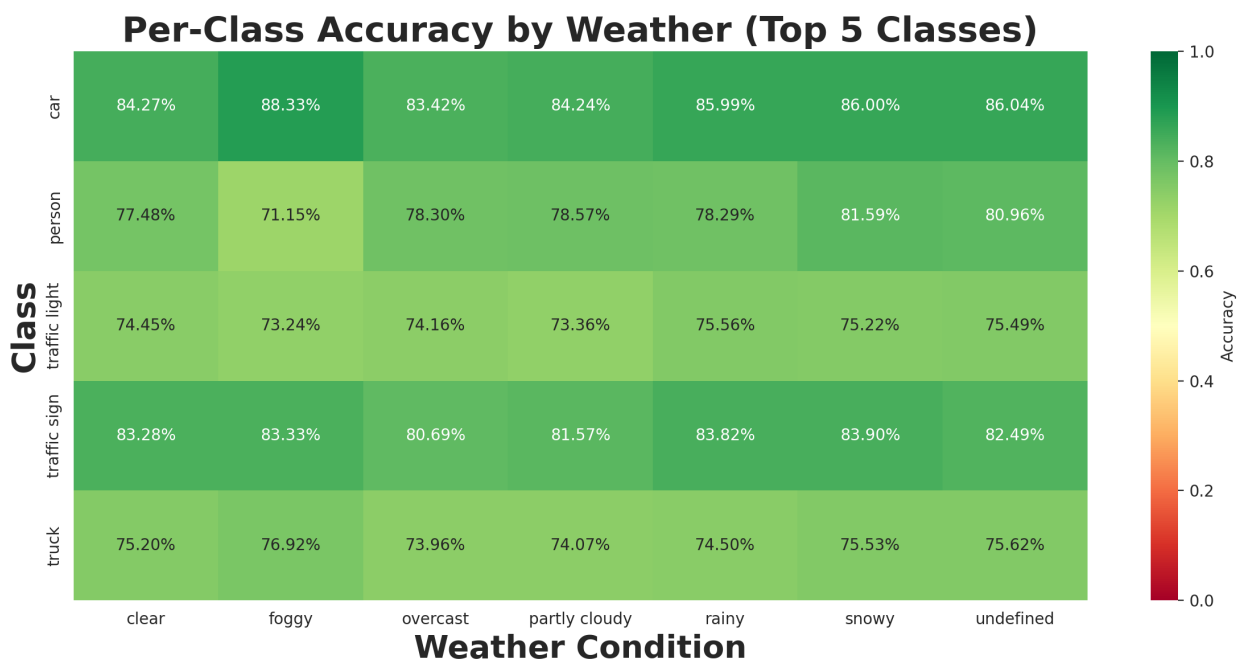
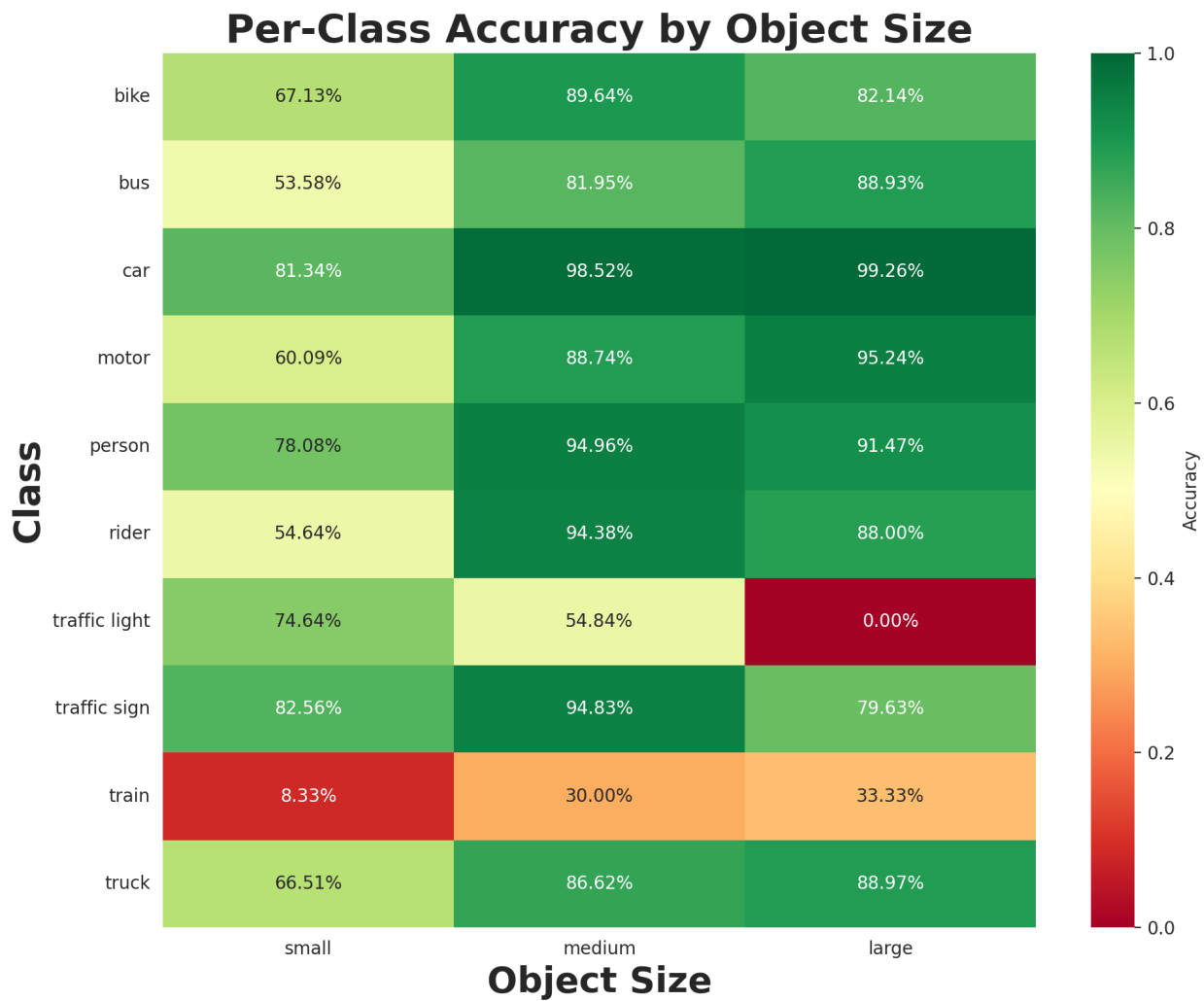


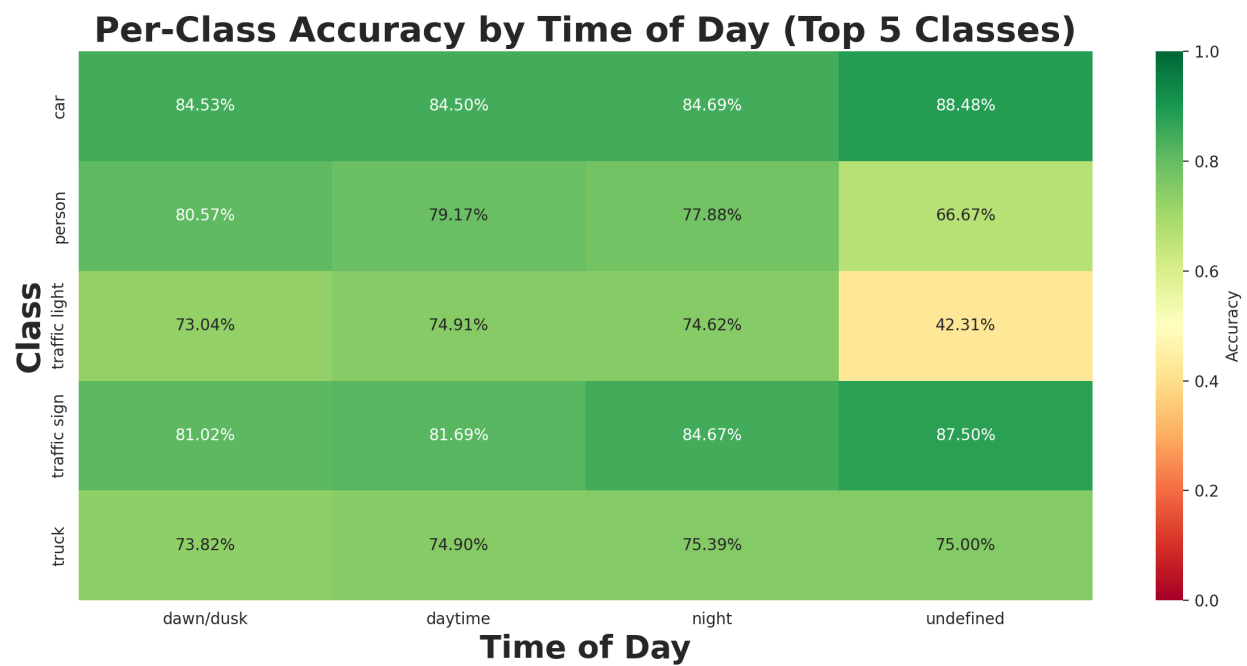
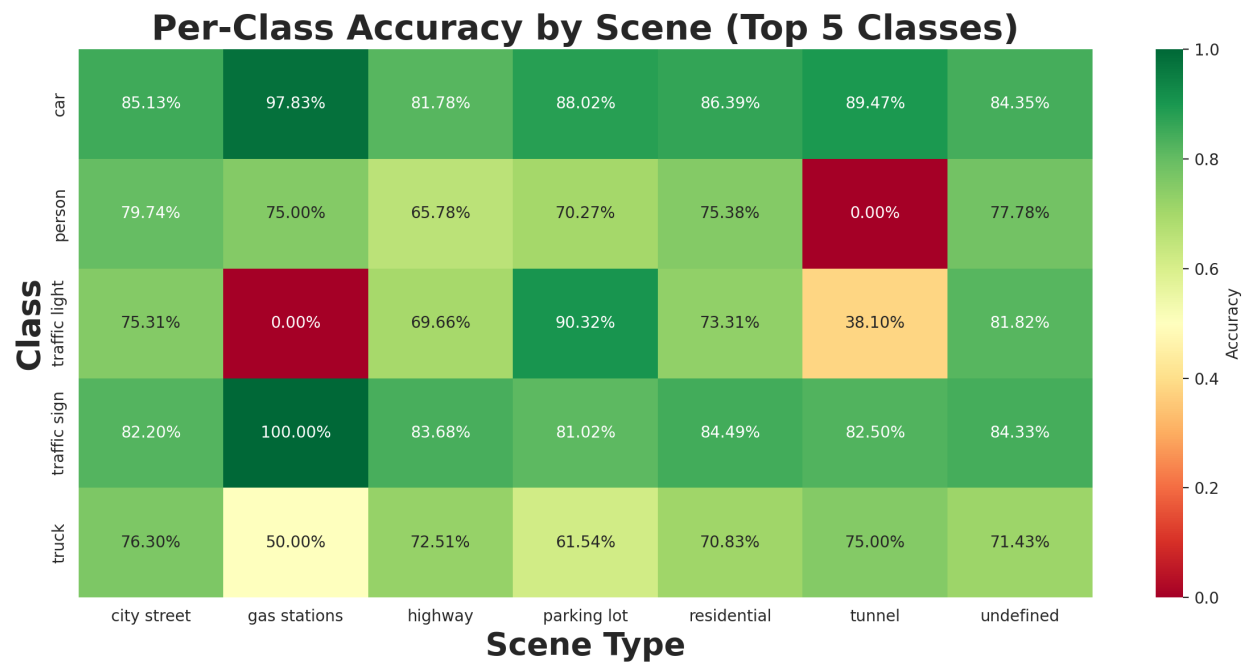


Accuracy by Object Characteristics



Per-Class Performance Analysis





Sample Predictions: Ground Truth vs Model

Sample #1: e6e9fe71-13d8ffac.jpg

Ground Truth Objects: 27 | Predicted Objects: 22

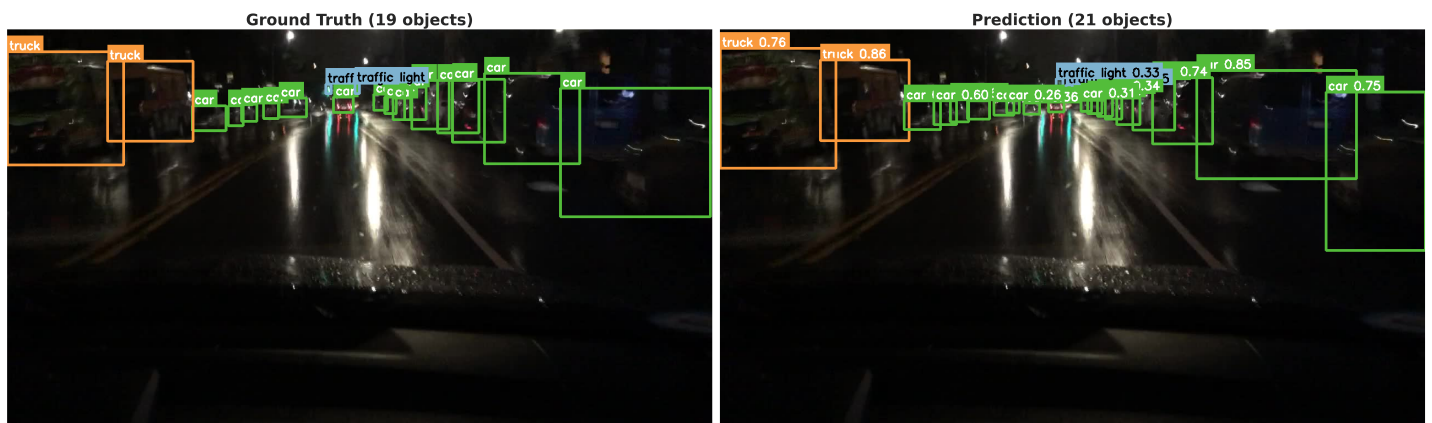
Weather: overcast | Scene: highway | Time of Day: daytime



Sample #2: d5959864-13a94fa2.jpg

Ground Truth Objects: 19 | Predicted Objects: 21

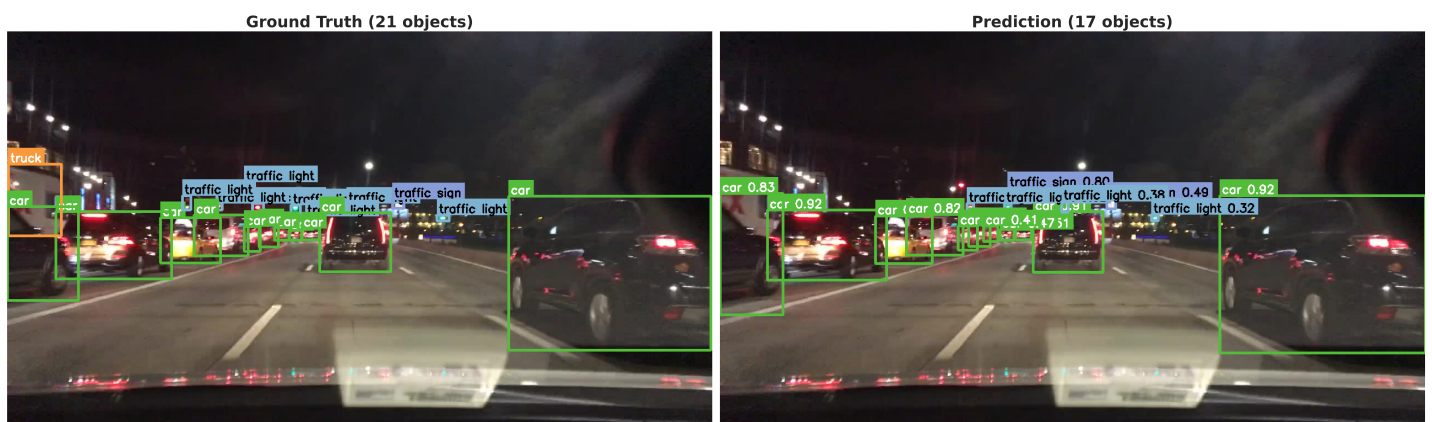
Weather: rainy | Scene: city street | Time of Day: night



Sample #3: cf32cfc4-0820fb1e.jpg

Ground Truth Objects: 21 | Predicted Objects: 17

Weather: clear | Scene: city street | Time of Day: night

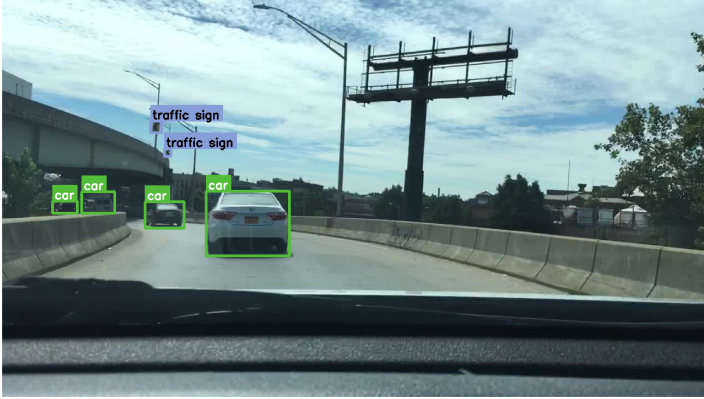


Sample #4: cca33354-a7713e9f.jpg

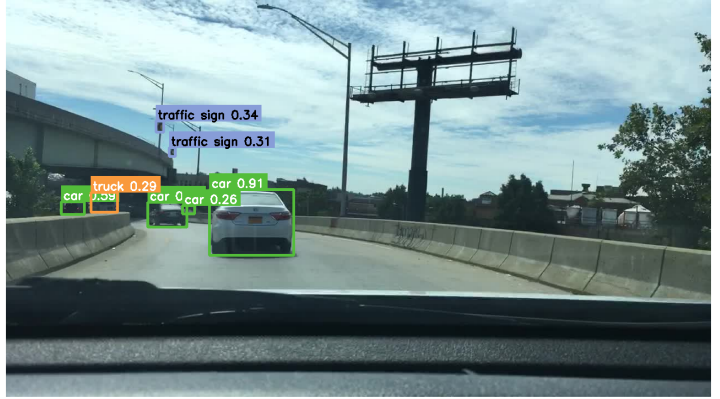
Ground Truth Objects: 6 | Predicted Objects: 7

Weather: overcast | Scene: highway | Time of Day: daytime

Ground Truth (6 objects)



Prediction (7 objects)

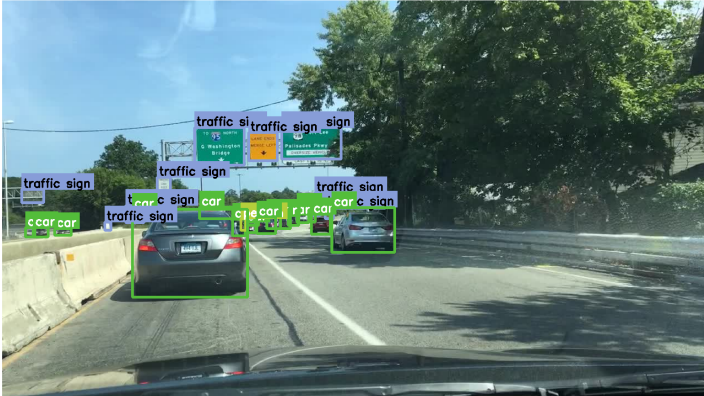


Sample #5: d2c4249f-4568f6ac.jpg

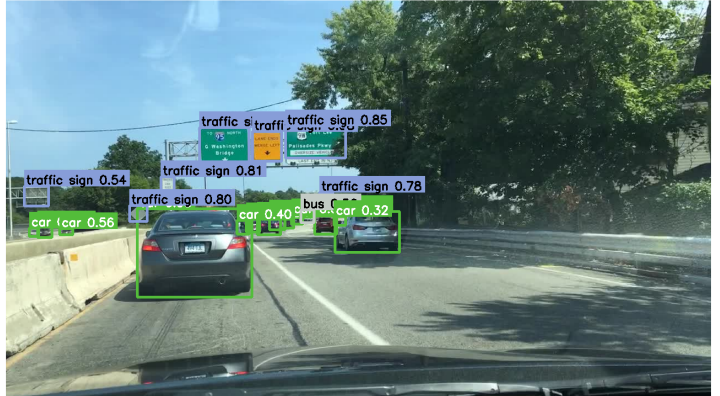
Ground Truth Objects: 27 | Predicted Objects: 21

Weather: clear | Scene: highway | Time of Day: daytime

Ground Truth (27 objects)



Prediction (21 objects)



Sample #6: cde381ce-f69c7cf2.jpg

Ground Truth Objects: 23 | Predicted Objects: 20

Weather: rainy | Scene: city street | Time of Day: night

Ground Truth (23 objects)



Prediction (20 objects)

